



Bias reduction using combined stain normalization and augmentation for AI-based classification of histological images

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ABSTRACT

Artificial intelligence (AI)-assisted diagnosis is an ongoing revolution in pathology. However, a frequent drawback of AI models is their propensity to make decisions based rather on bias in training dataset than on concrete biological features, thus weakening pathologists' trust in these tools. Technically, it is well known that microscopic images are altered by tissue processing and staining procedures, being one of the main sources of bias in machine learning for digital pathology. So as to deal with it, many teams have written about color normalization and augmentation methods. However, only a few of them have monitored their effects on bias reduction and model generalizability. In our study, two methods for stain augmentation (AugmentHE) and fast normalization (HENorm) have been created and their effect on bias reduction has been monitored. Actually, they have also been compared to previously described strategies. To that end, a multicenter dataset created for breast cancer histological grading has been used. Thanks to it, classification models have been trained in a single center before assessing its performance in other centers images. This setting led to extensively monitor bias reduction while providing accurate insight of both augmentation and normalization methods. AugmentHE provided an 81% increase in color dispersion compared to geometric augmentations only. In addition, every classification model that involved AugmentHE presented a significant increase in the area under receiving operator characteristic curve (AUC) over the widely used RGB shift. More precisely, AugmentHE-based models showed at least 0.14 AUC increase over RGB shift-based models. Regarding normalization, HENorm appeared to be up to 78x faster than conventional methods. It also provided satisfying results in terms of bias reduction. Altogether, our pipeline composed of AugmentHE and HENorm improved AUC on biased data by up to 21.7% compared to usual augmentations. Conventional normalization methods coupled with AugmentHE yielded similar results while being much slower. In conclusion, we have validated an open-source tool that can be used in any deep learning-based digital pathology project on H&E whole slide images (WSI) that efficiently reduces stain-induced bias and later on might help increase pathologists' confidence when using AI-based products.

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1. Introduction

Microscopic analysis relies on the observation of tissue slices using a brightfield microscope. In order to highlight biological structures, these slices are stained using hematoxylin and eosin (H&E). Actually, safran is also used in some laboratories. Hematoxylin stains cell nuclei in blue while eosin highlights cytoplasm and connective tissues in pink. Despite the use of the same chemical compounds, the actual slice colors strongly vary from one laboratory to another. Both dye manufacturers and particular staining protocols from different laboratories cause H&E color variability. Consequently, the origin of a slide can often be inferred from its stain characteristics. Besides, color intensity is altered by time, making old samples brighter and less contrasted than more recent ones. Altogether, these variations represent a frequent source of bias in digital pathology.

Machine Learning (ML) algorithms rely on the assumption that the training dataset accurately represents the diversity of real-world data. Unfortunately, it is hardly ever the case, especially in healthcare datasets in which bias is often induced by overrepresentation of frequent cases, a limited diversity of data sources and the small size of datasets. Such drawbacks can trigger overfitting, giving rise to ML models that are unable to generalize to real-world data. It is worth mentioning that in clinical research, patients' cohorts are statistically correlated to prognostic features on the basis of both inclusion and exclusion criteria. In short, the selection of patients can produce a major bias when building ML models.

A common workaround is to use data augmentation, which consists in modifying images using either geometric (rotation, translation, crop) or pixel level (color, contrast, luminosity, noise) transformations. Pixel-level transformations acting on images hue are commonly used to make model predictions less reliant on stain [1]. Unfortunately, these transformations often generate unrealistic output, creating noisy data that cannot exist in real world. Although little noise is welcome in ML as it helps models to better generalize, excessive noise can hinder their performance. Some attempts have been proposed to reduce stain-induced bias using stain deconvolution [2] to emulate real-world color variability [3] and create realistic images. Beside data augmentation, another procedure to reduce this bias is color normalization. While data augmentation artificially expands a dataset during training, normalization rather reduces variability at inference time. More specifically, normalization modifies colors of input images to make its stain similar to that of a reference dataset or image. This method can be used at training – combined with data augmentation – and at inference time. Even though it seems counterintuitive to concurrently use augmentation and normalization, the latter allows data to be concentrated around a reference point before being spread using augmentation. This enables normalized images to be integrated into the training color space at inference time. Several approaches to normalize images have been already reported. Most relied on the estimation of a color deconvolution matrix that sets a link between H&E stain and RGB space [2,4–6]. More recent methods used ML to identify morphological structures and associate them with one specific stain [7–9]. In addition, some authors used variational autoencoders [10] or generative adversarial networks (GAN) [11] to generate images, its colors matched with a reference stain [12–18]. Although these methods have been widely used in digital pathology, only a few studies described their actual performance regarding bias reduction [19,20].

In our study, we have developed innovative methods for stain augmentation (AugmentHE) and fast normalization (HENorm), that revisited Vahadane's method [6] to separate hematoxylin and eosin channels so as to better simulate real-world data. The impact of these methods on bias reduction has been studied and compared to other widespread techniques, therefore we used a multicenter dataset created for breast cancer histological grade assessment. Classification models were trained with single cohort images before assessing its performance on images coming from other cohorts. This setting led to extensively monitor bias reduction, thus providing accurate assessment of our augmentation and normalization methods.

2. Materials and methods

2.1. Cohort

A representative subset of available H&E slides from patients included in the PACS04 [21], PACS05 [22] and PACS08 [23] UNICANCER trials were used in our study. PACS04 was a phase III open-label randomized and multicentric study evaluating: (i) combined administration of docetaxel and epirubicin versus fluorouracil, epirubicin and cyclophosphamide (FEC)-100 in the treatment of non-metastatic lymph node-positive breast cancer patients; (ii) sequential addition of trastuzumab for [HER2 3+] or [HER2 2+ and FISH+] patients. PACS05 was a phase III randomized study of 4 versus 6 cycles of adjuvant FEC, in women with stage I breast cancer. PACS08 was a randomized open label multicentric phase III trial that evaluate the benefit of a sequential regimen associating FEC100 and Ixabepilone in adjuvant treatment of non metastatic poor prognosis breast cancer defined as triple negative tumor [HER2 negative – ER negative – PR negative] or [HER2 negative and PR negative] tumor in node positive or node negative patients. A representative surgical specimen paraffin block of the tumor was selected for each patient. H&E slide staining was centrally performed in a distinct laboratory for each trial, i.e. Institut Gustave Roussy (Villejuif, France), Centre Jean Perrin (Clermont-Ferrand, France) and Institut Claudius Regaud (Toulouse, France) for PACS04, PACS05 and PACS08 respectively. Histological grade was based on the evaluation of 3 items : tubule formation, nuclear pleomorphism and mitotic count as described by C.W. Elston and I.O. Ellis in 1991 [24]. Each item is scored between 1 and 3. The sum of the 3 scores defines the histological grade: 3, 4 or 5 = grade I; 6 or 7 = grade II; 8 or 9 = grade III. These three trials were approved by regulatory and ethics authorities of all participating centers (ClinicalTrials.gov identifier, NCT00054587, NCT00055679 and NCT00630032 for PACS04, PACS05 and PACS08, respectively). The project complies with the reference methodology MR-004 and has been registered in the public directory of the Health Data Hub (F20220930101746). The methods were performed in accordance with relevant guidelines and regulations and approved by CNIL and the Projects Review and Ethical Committee of the Institut Claudius Regaud - IUCT-Oncopole. Owing to the retrospective nature of this study and to the respect of the reference methodology MR-004, the ethics committee granted a waiver of informed consent for the included participants.

2.2. Whole Slide Images (WSI) acquisition

H&E slides were digitized using a Panoramic 250 Flash II digital microscope (3DHISTECH, Budapest, Hungary) at 20× magnification to achieve a scan resolution of 0.24 $\mu\text{m}/\text{pixel}$.

2.3. AugmentHE: Histology-specific method for stain color augmentation

AugmentHE was specifically designed for stain color augmentation. It relies on the assumption that stain color variability of H&E slides is related to (i) the dye composition might differ from one supplier to another. It also leads to significant differences in stain. For example, eosin hue ranges from intense pink to light orange. In addition, some laboratories use safran adding a yellowish hue of collagen bundles; (ii) the slide preparation protocol depends on slice thickness or dye amount, it might also cause variability in luminosity, contrast or color saturation; (iii) the slides oldness when stains tend to become less contrasted and saturated through time. This fading effect can be troublesome for retrospective cohorts with several years follow-up. Therefore, our augmentation method should affect two variables: color produced by both hematoxylin and eosin and color concentration that might be altered by tissue thickness or dye amount. Using Vahadane's method [6] thanks to StainTools library [25], given $I \in \mathbb{R}^{3 \times h \times w}$ (where h and w are the image height and width respectively) the image in

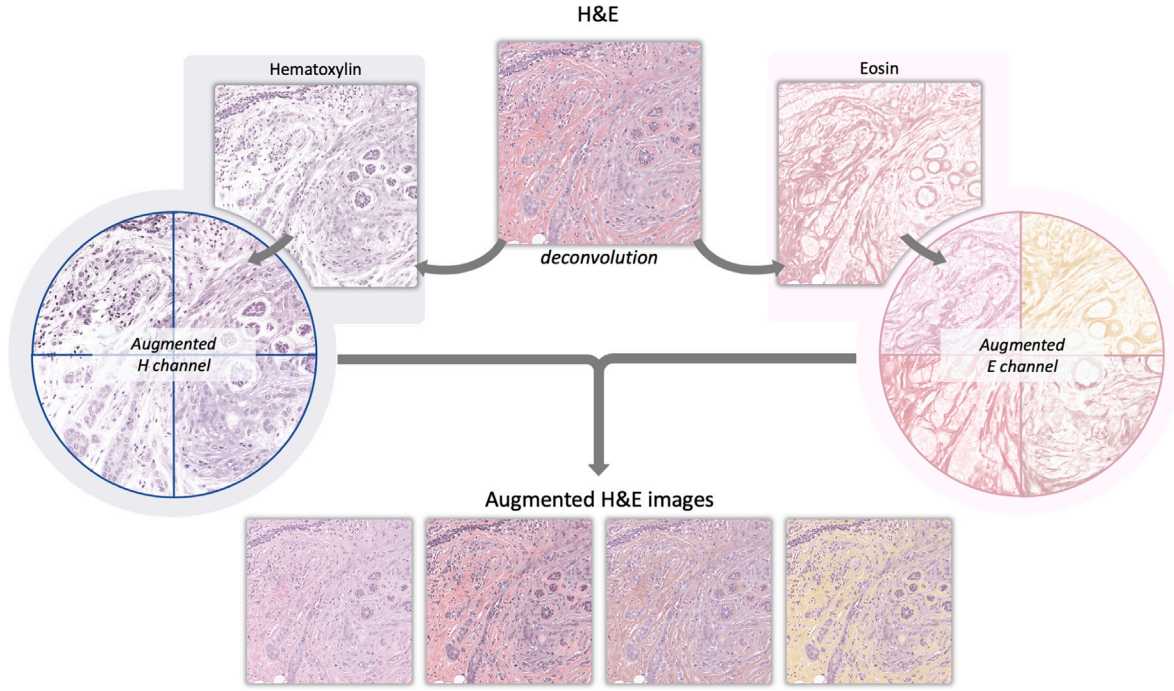


Fig. 1. Illustration of AugmentHE method. Hematoxylin and eosin channels are first extracted from the input image. They are then randomly modified and recombined to create a restained image. Four examples of potential output images are displayed herein.

RGB space, a stain color matrix $W \in \mathbb{R}^{3 \times 2}$ and a stain density matrix $H \in \mathbb{R}^{2 \times hw}$ were computed such that:

$$I = \exp(-WH) \quad (1)$$

The W matrix columns represent colors in RGB space that pure hematoxylin and eosin produce. The H matrix represents the amount of hematoxylin and eosin each pixel contains. This implies that modifying W artificially changes the dye composition, while modifying H changes dye concentration. We therefore sampled $\alpha_W \in [1 - \alpha_W^{lim}, 1 + \alpha_W^{lim}]^{3 \times 2}$, $\beta_W \in [-\beta_W^{lim}, \beta_W^{lim}]^{3 \times 2}$, $\alpha_H \in [1 - \alpha_H^{lim}, 1 + \alpha_H^{lim}]^2$, $\beta_H \in [-\beta_H^{lim}, \beta_H^{lim}]^2$ (with $\alpha_W^{lim}, \beta_W^{lim}, \alpha_H^{lim}, \beta_H^{lim} \in \mathbb{R}$) from uniform distributions. We thus defined $W' \in \mathbb{R}^{3 \times 2}$ and $H' \in \mathbb{R}^{2 \times hw}$ such that:

$$W' = \alpha_W \cdot W + \beta_W \mathbb{I}_{3,2}, \text{ with } (\cdot) \text{ denoting element-wise product.}$$

$$H' = \begin{cases} \alpha_H H + \beta_H \mathbb{I}_{2,hw} & \text{where } H > k \\ H & \text{otherwise} \end{cases} \quad (2)$$

Using a threshold value k for concentration levels ensures that background remains unchanged. We set $k = 0.2$ in all experiments. We then defined a new altered image $I' \in \mathbb{R}^{3 \times hw}$ as:

$$I' = \exp(-W'H') \quad (3)$$

I' is identical to I except for its randomly generated stain. This enabled us to represent a very large diversity of color stains from a single source, which helped us to reduce the associated bias. The whole process is illustrated in Fig. 1.

2.4. HEnorm: deep learning-based method for H&E images normalization

2.4.1. Normacolor dataset

The Normacolor dataset is a 700 fully-anonymized WSI collection of breast tissue slices gathered in the Pathology Department of the University Cancer Institute of Toulouse - Oncopole. This WSI acquisition was done in a short period of time in order to avoid any substantial change in specimen handling, slicing technique and staining protocol. Therefore, this collection was composed of homogeneously stained WSI breast tissue. These WSI were then divided into contiguous 1024×1024 -pixel patches filtered to remove background, creating

more than 64,000 images. This dataset was then split into training and validation sets. Tissue samples were collected and processed at the “CRB Cancer des Hôpitaux de Toulouse”, following ethical procedures (Declaration of Helsinki). In accordance with French law, the “CRB Cancer des Hôpitaux de Toulouse” Cancer collection was reported to the Ministry of Higher Education and Research (DC 2009–989). A transfer agreement (AC-2008-820) was obtained following approval by the ethics committees.

2.4.2. Model

HEnorm is a deep learning (DL)-based normalization method that is trained using Normacolor dataset and AugmentHE. At training time, images were randomly re-stained using AugmentHE before being transferred into H&E space using a scikit-image [26] implementation that we adapted for PyTorch [27]. The H channel has been used alone as input to a U-Net model [28] expected to reconstruct the original non-augmented H&E image, since the most relevant features of tissue architecture lie in this channel. As a matter of fact, hematoxylin counterstaining is widely used in routine practice since it allows an accurate appreciation of tissue architecture mainly in tumoral tissues in which tumor cellularity allows pathologists to capture these patterns. Nevertheless, there might be some exceptions such as collagen-rich tissues or some parasitic infections in which key image patterns lies in the eosin channel [29]. Our method forced a full E channel restaining that often exhibits most of the stain variability. The whole model is illustrated in Fig. 2.

We have been using both PyTorch [27] and Pytorch Lightning (<https://www.pytorchlightning.ai/>) libraries. FastAI's [30] DynamicUnet (<https://docs.fast.ai/vision.models.unet.html>) has been implemented with custom CGR_5_32_4 encoder (Supplementary Figure 1). Four-image batches have been applied and backpropagation has been performed using Mean Squared Error (MSE) loss with AdamW optimizer [31]. The training process practically followed two steps:

1. 10 epochs with cosine-annealed OneCycle scheduler [32] with a maximum learning rate of 2×10^{-4} and a weight decay of 10^{-3} ;

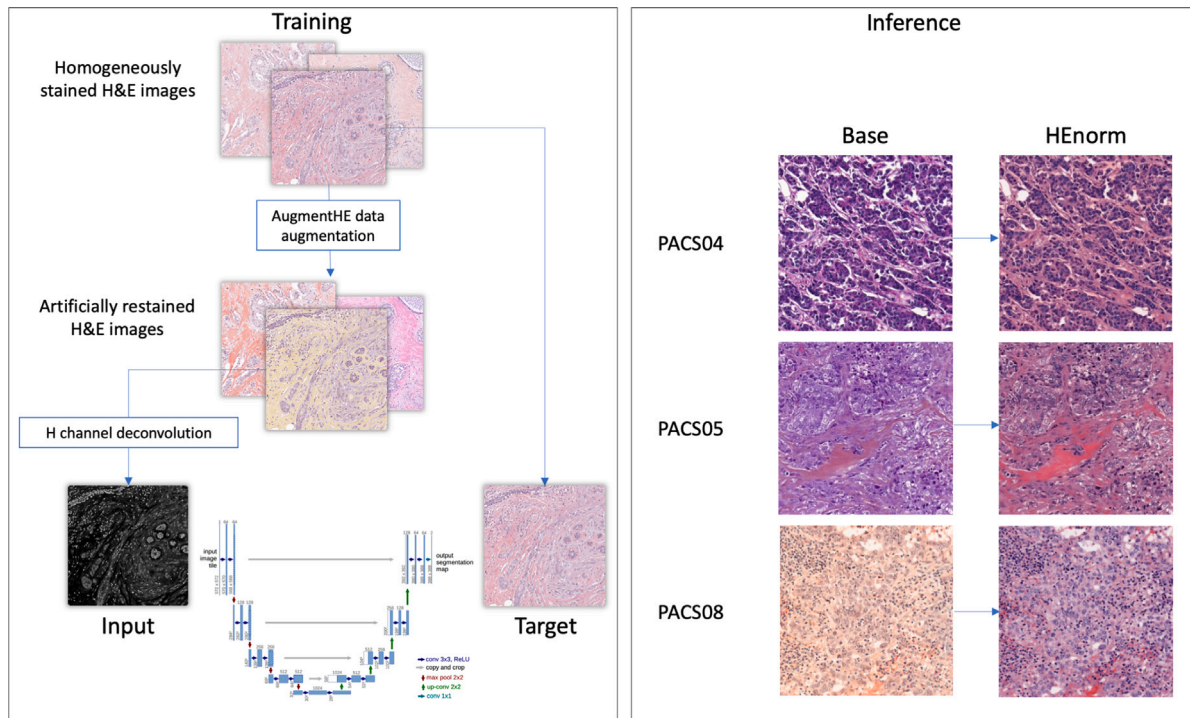


Fig. 2. Illustration of HEnorm method. Images are first augmented using AugmentHE and their hematoxylin (H) channel are extracted. These H channels then went through a U-Net model in order to reconstruct the original non-augmented images. Some results are included on the right.

2. 150 epochs with reduced learning rate on plateau scheduler with early stopping, an initial learning rate of 10^{-3} and a weight decay of 2×10^{-4} .

The total amount of training time was about 600 h using a single Tesla P100 PCIe 16 GB GPU.

2.5. Effectiveness measurement of AugmentHE and HEnorm on bias reduction

In order to measure the effectiveness of AugmentHE and HEnorm on bias reduction we focused on a single task: histological grade classification of breast cancer. We chose to focus on grade I and grade III distinction since we have essentially wanted to monitor bias reduction on a straightforward and intuitively simple classification task. The classification of grade I versus grade III BC has already been clinically relevant [33]. Particularly, our goal was to discriminate between histological grades I and III on the basis of H&E WSI. Our aim was also to test whether AugmentHE and HEnorm improved the generalization ability of a model trained on biased data, instead of reaching high accuracy since automated grade assessment remains challenging.

2.5.1. Data

To that end, our dataset was split up depending on the source of the images cohort. As it contained most of these images (Supplementary Table 1), PACS04 was used for training, whereas PACS05 and PACS08 were kept for biased test. Ten percent of PACS04 images were put aside for validation while the remaining 90% were divided into 5 folds for k-fold cross-validation. Tissue detection was then computed using thresholding in the $L^*a^*b^*$ space on WSI thumbnails, before extracting contiguous 1024×1024 -pixel patches in foreground regions. This resulted in about 420,000 PACS04 patches, being 80,000 histological grade I and 340,000 histological grade III respectively. In order to face this data imbalance, we randomly undersampled grade III images during training step, showing up a difference of less than 3% in class populations. This is to say that only 40,000 images were used at each epoch. During the inference step, 20,000 PACS05 patches (3000 grade

I and 17,000 grade III) and 28,300 PACS08 patches (800 grade I and 27,500 grade III) were applied.

2.5.2. Model

We tried different combinations of bias reducing techniques, all including at least geometric augmentations, in order to compare them between each other. Our baseline used only geometric augmentations such as flips and transpositions (noted as basic). We compared it to a classical augmentation method called RGB shift which applies a random offset to the values of each RGB channel (noted as aug_classic). We then evaluated both AugmentHE and HEnorm separately (respectively noted as augHE and HEnorm) and both combined (noted as HEnorm_augHE). In order to compare HEnorm to the state of the art, we evaluated combined AugmentHE and Vahadane's normalization method [6] (noted as Vahadane_augHE). For each of these combinations, we trained 10 ResNet50 [34] classifiers, using two series of 5-fold cross-validation. Backpropagation was performed using binary cross-entropy loss and AdamW optimizer [31]. We used a One Cycle [32] policy with a maximum learning rate of 2×10^{-4} and a weight decay of 10^{-3} . Training was performed using batches of 8 images for 10 epochs, with a single Tesla P100 PCIe 16 GB GPU.

3. Data availability

The WSI from the three cohorts PACS04, PACS05 and PACS08 are the property of Unicancer. Unicancer certifies that it will share de-identified individual data that underlie the results reported under the following conditions: - Unicancer will grant access to all study data to all requestors in the field of cancer research, upon written detailed request sent to DAC-OS2006@unicancer.fr - Unicancer will grant access to editor's reviewers to study documents and to the data required for independent verification of the published results. Editor guarantees that mandated reviewers are committed to use the data transferred for the sole aim to reviewing produced results, under strict conditions warranting security and confidentiality of the data.

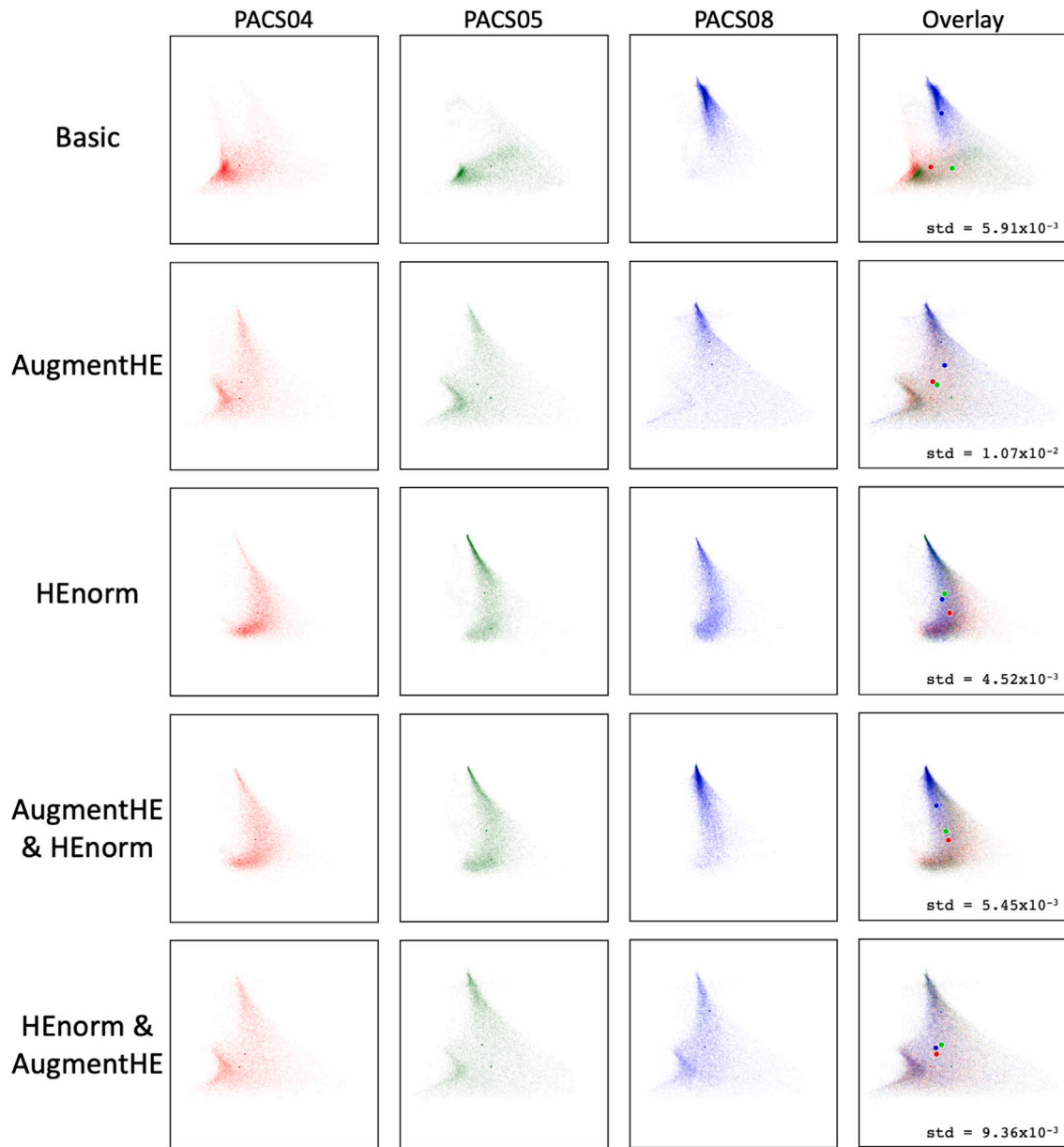


Fig. 3. 2D representation of color dispersion when using different sets of transformations. Points are 2D orthogonal projections of normalized hue histograms from the different cohorts. The projection plane contains the centroids from the three cohorts with no transformation applied. These centroids are highlighted on the overlay column (for comparison, centroids from the basic setup are also highlighted). Average standard deviation across all histogram bins is also given.

4. Code availability

Source code for HEnorm and AugmentHE is available at <https://github.com/schwobr/HEnorm>. It includes pretrained weights for HEnorm on 1024×1024 images at level 1 ($0.5 \mu\text{m}/\text{px}$).

5. Results

5.1. PACS04, PACS05 and PACS08 WSI exhibited a major color bias

On the one hand, the cohorts had been previously stained in distinct central laboratories, resulting in a color palette belonging to each of them. On the other hand, inclusion criteria of these clinical trials were disparate. Indeed, PACS04 and PACS05 patients harbored rather good prognoses whereas PACS08 patients showed less favorable outcomes. Consequently, a direct link between color palette and aggressiveness of tumors appeared. To address the question whether both data augmentation and color normalization might reduce this bias, we have

randomly selected a subset of grade I and grade III tumors following the overall cohort proportions. Thus, we have included 128 grade I and 436 grade III tumors from PACS04, 9 grade I and 221 grade III tumors from PACS05, 3 grade I and 328 grade III tumors from PACS08 respectively.

5.2. AugmentHE pragmatically increased dataset color dispersion

The purpose of color augmentation is to better cover the variability of real data. Basically, regarding microscopic images, a special attention must have been paid to preserve biological information. AugmentHE achieved this goal by generating credible pathological images (Fig. 1) while color dispersion successfully increased. Indeed, we assessed the normalized hue histograms of all images and computed their standard deviation for each bin. The average standard deviation (ASD) increased from 5.91×10^{-3} with only geometric augmentations to 1.07×10^{-2} with AugmentHE (Fig. 3), providing a 81% rise. We included the 2D projections of these hue histograms on a plane defined by the centroids

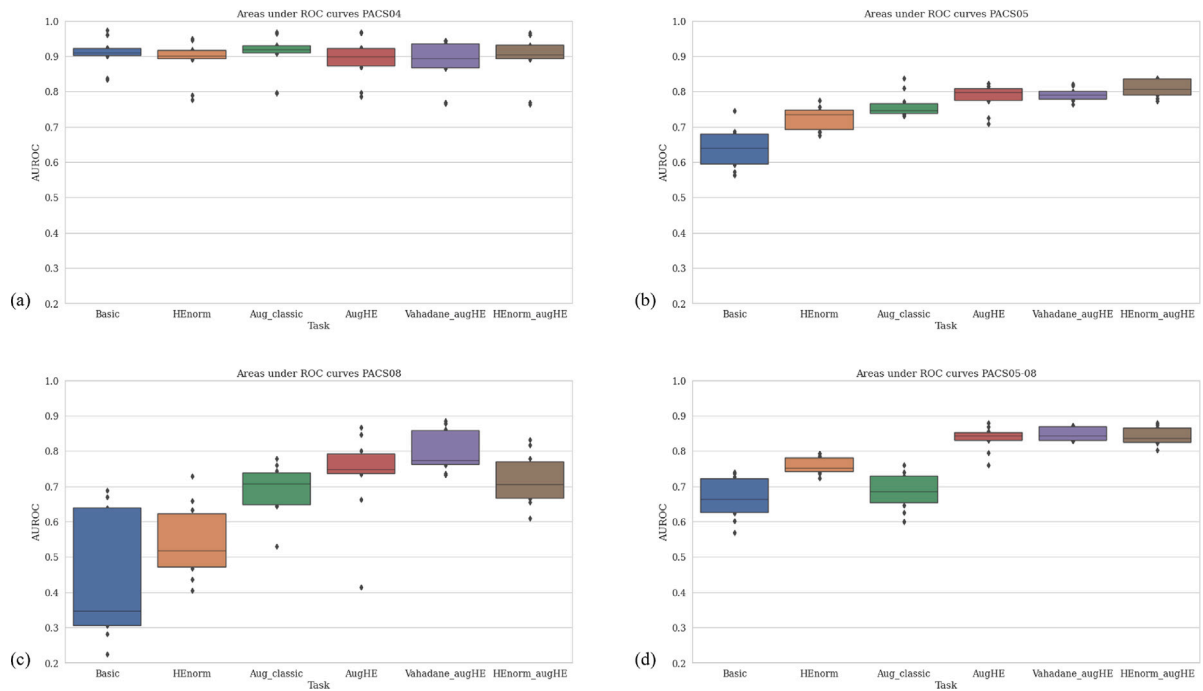


Fig. 4. The areas under receiver operating characteristic curves (AUC) from 10 models trained on PACS04 images for each task on (a) PACS04 validation set (b) PACS05 whole dataset (c) PACS08 whole dataset and (d) combined PACS05 and PACS08 datasets. Augmentation settings are labeled as follows: Basic = only geometric augmentations (rotation, flip, transpose); AugHE = Basic + AugmentHE; Aug_classic = Basic + RGB shift. In addition, both HEnorm and Vahadane's methods [6] were combined with AugmentHE. All average AUC values can be found in Supplementary Table 7.

of the 3 cohorts (with no augmentation applied), in Fig. 3. These point clouds clearly confirmed the increase in data dispersion when using AugmentHE. Besides, the centroids of the cohorts were brought closer together by these augmentations, which proved that AugmentHE made the cohorts color distributions harder to distinguish.

5.3. HEnorm efficiently normalized stain while preserving image structure

HEnorm managed to normalize colors while preserving tissue structure at a low computational cost. To compare, it took 30 ms to normalize a 1024×1024 -pixel image using HEnorm, being $3.5\times$ faster than Reinhard's [4], $50\times$ faster than Macenko's [5] and $78\times$ faster than Vahadane's [6] methods respectively (Supplementary Table 2). Despite HEnorm presented a slightly less precise image structure preservation than the previously mentioned methods, the structural similarity (SSIM) and Pearson correlation coefficient (PCC), ranging from 0 to 1, attained 0.891 and 0.889 respectively (Supplementary Table 2).

Concerning reduction of differences in color distribution between cohorts, Fig. 3 shows that HEnorm decreased the ASD of normalized hue histograms by 25%. Furthermore, point clouds are visually overlapping, which is illustrated by bringing closer centroids of these 3 cohorts.

5.4. Combination of both AugmentHE and HEnorm contributed to bias reduction

When it comes to ML algorithm training, cohorts containing images from different origins should be indistinguishable in order to avoid any bias. To that end, it is common to chain normalization and augmentation, also leading to wide enough data dispersion so as to prevent overfitting. Intuitively, applying these transformations in reverse order would be counterproductive as normalization should cancel out stain augmentation. This is confirmed in Fig. 3, where point clouds obtained when using AugmentHE before HEnorm looked alike those obtained using HEnorm only. Furthermore, ASD only increased by 21% when adding AugmentHE before HEnorm. In contrast, when

using AugmentHE only, ASD was 137% higher than with HEnorm. Conversely, adding HEnorm before AugmentHE only decreased ASD by 12%, with the added benefit of a near perfect overlap between point clouds and its centroids. Consequently, the use of HEnorm followed by AugmentHE looked like the optimal combination for training step. Practically, the main goal of our study was to monitor generalizability of models trained on biased datasets applying different combinations of data augmentation and normalization. On the one hand, we determined that augmentation and normalization caused no significant decrease in validation area under receiving operator characteristic curve (AUC) on PACS04 (Supplementary Table 3) regarding AUC values ranging from 0.88 ± 0.06 for Vahadane_augHE to 0.91 ± 0.04 for basic. On the other hand, augmentation and normalization significantly increased AUC on test data from combined PACS05 and PACS08 (Fig. 4 and Supplementary Table 6). More specifically, every combination that involved AugmentHE caused a significant increase in AUC over basic or aug_classic. According to this, AUC ranged from 0.67 ± 0.06 and 0.69 ± 0.05 for basic and aug_classic respectively, and reached up to 0.83 ± 0.03 , 0.84 ± 0.02 and 0.85 ± 0.02 for augHE, HEnorm_augHE and Vahadane_augHE respectively. Similarly, only models trained without AugmentHE showed significant decrease in AUC from validation to test steps, with a 25% reduction on average (Supplementary Table 8). More technically, loss function values are presented in Supplementary Figure 2.

Looking at results on PACS05 and PACS08 separately, some questions raised concerning the effectiveness of each combination. In particular, Vahadane_augHE showed no significant increase in AUC over aug_classic or HEnorm on PACS05, whereas HEnorm_augHE had no significant increase in AUC over aug_classic on PACS08 (Supplementary Tables 4 and 5). Strikingly, all methods showed a significant decrease in AUC from validation to test steps on separated PACS05 and PACS08 with the exception of Vahadane_augHE on PACS08 (Supplementary Table 8). A noticeable difference thus appeared between results obtained when combining PACS05 and PACS08 compared to those obtained on them separately. For instance, AUC coming from HEnorm_augHE experiment was 0.85 ± 0.02 on pooled PACS05 and PACS08, whereas

it was only 0.82 ± 0.02 on PACS05 and 0.73 ± 0.07 on PACS08. This difference can be explained by the massive imbalance between grade I and grade III in PACS05 and PACS08. Indeed, even with a model that would classify all PACS08 images as grade III, the false positive rate on combined PACS05 and PACS08 cohorts would remain rather low as grade I tumors of PACS08 only represent 25% of all grade I images. On the contrary, when working on PACS08 alone, predicting all images to be grade III implies having a false positive rate of 100%. It is the same case for PACS05 to a lesser degree, as it contains a higher amount of grade I images than PACS08.

6. Discussion

Two innovative approaches for data augmentation (AugmentHE) and color normalization (HENorm) have been developed in our study, providing an end-to-end pipeline for histological images preparation within DL projects. Respective impacts of AugmentHE and HENorm have been assessed and compared to widely used techniques previously described in the literature [4–6]. Thus, the most efficient combination of both AugmentHE and HENorm regarding bias reduction has been established herein. As pathologists, we developed AugmentHE as a pragmatic color augmentation method based on our daily practice in laboratories. This method takes advantage of our tangible knowledge of the staining process as a broad source of color variability. It was also greatly inspired by Tellez et al. [19] as their HED augmentation helped us to translate our knowledge regarding the staining process into an actual augmentation pipeline. Thanks to it, we generated quite convincing and realistic histological images compared to those produced using color augmentation methods based on non-specific approaches such as RGB or hue shifts. More importantly, a significantly better bias reduction was obtained with AugmentHE compared to classical methods. Furthermore, HENorm appeared to be a fast and reliable method for color normalization that considered the major role of eosin composition in color variation of H&E slides. Actually, eosin is frequently associated with orange or yellow dyes that drastically modify the hue of H&E images from one laboratory to another. In addition, eosin fades with time and light exposure rather than hematoxylin, leading to increased color variability. On the contrary, hematoxylin is shown to be more stable and harbor most of the characteristics about tissue structure that can be accurately used as a canvas for H&E image reconstruction. Accordingly, HENorm is a DL-based H&E generator from hematoxylin input images providing normalized output images. In terms of results, HENorm efficiently decreased color dispersion from one cohort to another and was much faster than every other method herein tested [4–6]. Interestingly, according to generalizability, our method significantly reduced bias when combined with AugmentHE concerning PACS05 images. However, our model behaved differently regarding PACS08 cohort in which Vahadane's method combined with AugmentHE provided the best bias reduction results. Concerning results from color normalization, neither HENorm nor Vahadane's method alone could fully reach bias reduction expectations. Therefore, it clearly appeared that color normalization could be regarded as a fine tuning tool applied at the end of ML model development to further reinforce generalizability. Besides, results were significantly better on combined PACS05 and PACS08 than on each cohort alone. It is a clear illustration that good results can hide bias if they are not properly analyzed through more precise lenses. We assessed both when and how to apply AugmentHE and HENorm by exploring every possible combination in order to provide the most optimal sequence in terms of bias mitigation. We therefore showed that the best way to bring the hue of cohorts closer is to firstly apply normalization and then data augmentation, thus it considerably reduced stain-induced bias. Besides these satisfying results, several options should still be considered as potential enhancements of our pipeline. Firstly, AugmentHE might benefit from an even more realistic modelization of color variations between laboratories. In fact, we chose to enlarge color dispersion on the basis of our

laboratory staining color space. Another option could have been to directly represent real world color variation for which unfortunately there is still no precise knowledge available. In practice, defining such a color space would require gathering a large multicenter dataset as an acceptable representation of real-world variability. Secondly, in terms of color normalization, Tellez et al. showed that the mix of several normalization methods can help to reduce bias [19]. The latter study corroborates our results showing that there is no universal normalization method that is effective on all datasets. Thirdly, we identified potential improvements on HENorm method that would rely on adversarial learning techniques. Indeed, HENorm was trained using a simple MSE loss, which has been perfectly designed for image reconstruction despite not being state-of-the-art. Therefore, this reconstruction may benefit from adversarial loss, as generated images should not be distinguishable from the original ones [13,35]. Actually, such adversarial techniques have already been described in the literature [12–15,18] and demonstrated better results than classical methods did [36]. We admit this is one out our study's main limitations and the impact of those more recent approaches on classifier performances, combined or not with AugmentHE, should be addressed in future works. Another aspect that could be considered as a limitation is the choice to present classification performance only at patch level, without aggregating results at WSI level. Indeed, the spatial heterogeneity of breast cancers may account for variations in grade at patch level that should be considered when aggregating data at WSI level [37]. However, it is important to emphasize that the primary objective of our study was to monitor bias reduction through some normalization and augmentation tools, including ours. Consequently, performances evolution at patch level was sufficient to assess the impact of various combinations of data augmentation and normalization. Furthermore, recent papers have moved away from the concept of grade to directly classify tumors based on the presence or absence of early BC recurrence [38]. However, we speculate that these approaches would also benefit from the use of bias reduction tools, such as those proposed herein. Concerning the fact that our work is fully based on BC samples, both AugmentHE and HENorm can be applied to other cancer types. On the one hand, AugmentHE is already applicable to every kind of H&E histological images. On the other hand, HENorm could either work on other types of cancer without being modified, or might need an additional training based on targeted tissue. Moreover, we remain uncertain that HENorm would be efficient in rare cases in which key image patterns lie in the eosin channel, such as collagen-rich tissues or some parasitic infections. Further research should be carried out in order to validate HENorm in these cases.

7. Conclusion

Our work is thus part of a general breakpoint in ML history where there is a growing interest in the study of bias after a long period of time where proof of concepts have been at the forefront of ML research. Indeed, we quantified stain-induced bias in a multisite dataset as well as the effectiveness of several augmentation and normalization methods, including our own. According to the fact that nowadays datasets are hardly ever optimally designed for ML studies, we propose a pipeline that helps mitigating bias in ML models for histological images. Our pipeline should also serve as a baseline for future proof of concepts that need a ready-to-use bias-reduction method as well as new studies on bias in digital pathology. Indeed, thanks to its versatility, our method is not only designed for color-bias reduction in breast cancer studies but can also adapt to pan-cancer studies.

CRedit authorship contribution statement

Camille Franchet: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Robin Schwob:** Writing – review & editing, Writing – original draft,

Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Guillaume Bataillon**: Validation. **Charlotte Syrykh**: Validation. **Sarah Péricart**: Validation. **François-Xavier Frenois**: Visualization, Software. **Frédérique Penault-Llorca**: Resources, Funding acquisition. **Magali Lacroix-Triki**: Resources, Funding acquisition. **Laurent Arnould**: Resources, Funding acquisition. **Jérôme Lemonnier**: Resources, Data curation. **Jean-Marc Alliot**: Conceptualization. **Thomas Filleron**: Validation, Methodology. **Pierre Brousset**: Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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